

Program : Diploma in Robotic Process Automation	
Course Code : 5307	Course Title: Introduction to Machine Learning lab
Semester :5	Credits: 1.5
Course Category: Program Core	
Periods per week: 3(L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- Learn to perform data extraction and preprocessing using python tool kits.
- Demonstrate the working of classification and regression algorithms.
- Demonstrate the working of clustering algorithms
- Understand and analyze the performance of neural networks in classification.
- To obtain practical knowledge in real world problems.

Course Prerequisites:

Topic	Course code	Course name	Semester
Vector Calculus, Linear Algebra, Probability	1002 & 2002	Mathematics I &II	1 & 2
Python programming knowledge	3301	Problem solving using python	3

Course Outcomes:

On completion of the course student will be able to:

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Understand the features of machine learning to apply on real world problems using python programming language.	8	Applying

CO2	Understand the different types of Classification and regression algorithms.	15	Applying
CO3	Design and evaluate the unsupervised models through python in built functions.	8	Applying
CO4	Analyze the concepts of perceptron and SVM and implement using python packages.	12	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3	2	2	3	3		
CO2	3	2	2	3	3		
CO3	3	2	2	3	3		
CO4	3	2	2	3	3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the features of machine learning to apply on real world problems using python programming language.		
M1.01	Study of Python Basic Libraries such as Statistics, Math, Numpy and Scipy	4	Applying
M1.02	Study of Python Libraries for ML application such as Pandas and Matplotlib	4	Applying
CO2	Understand the different types of Classification and regression algorithms		

M2.01	Preprocess the dataset	3	Applying
M2.02	Implementation of Bayesian classifier and analyzing the classification performance.	4	Applying
M2.03	Implementation of simple and linear regression using python built in functions.	4	Applying
M2.04	Implementation of multiple linear regression using python built in functions.	4	Applying
	Series Test – I	1	
CO3:	Design and evaluate the unsupervised models using python in built functions.		
M3.01	Implement K-means clustering algorithm to cluster a set of data stored in a csv file	4	Applying
M3.02	Implement EM clustering algorithm to cluster a set of data stored in a csv file	4	Applying
CO4	Analyze the concepts of perceptron and SVM and implement using python packages.		
M4.01	Implement perceptron classifier using python sklearn	4	Applying
M4.02	Implement SVM classifier using python sklearn	4	Applying
M4.03	Performance analysis of classification algorithms on a specific dataset.	4	Applying
	Series Test – II	1	

**** - Micro Project**

Students are expected to do a micro project in machine learning during the course for the purpose of continuous evaluation. This experiment shall be included in the bona-fide record. Example: Develop program such as

- Classification of Mushroom dataset (UCI repository)
- House price prediction (Regression)
- K-Means clustering (Iris dataset)

Text / Reference

T/R	Book Title/Author
T1	Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006
T2	Ethem Alpaydın, Introduction to Machine Learning (Adaptive Computation and Machine Learning), MIT Press, 2004.

Online Resources

Sl.No	Website Link
1	https://www.javatpoint.com/machine-learning
2	https://www.toptal.com/machine-learning/machine-learning-theory-an-introductory-primer
3	https://www.tutorialspoint.com/machine_learning/index.htm
4	https://ml-course.github.io/master/labs/Lab%201%20-%20Tutorial
5	https://www.geeksforgeeks.org/introduction-machine-learning-using-python/

List of Experiments:

1. Write a python program to demonstrate the working of Numpy functions.
2. Write a python program to demonstrate the working of math library functions.
3. Write a python program to read data from a csv file and convert it to pandas data frame.
4. Write a python program to plot graphs using Matplotlib library.
5. Write a python program to preprocess the dataset by substituting mean, mode or median for missing data.
6. Write a python program to implement a Bayesian classifier and analyze the classification performance.
7. Write a python program to implement simple linear regression using built in functions.
8. Write a python program to implement multiple linear regression using built in functions.
9. Implement K-means clustering algorithm to cluster a set of data stored in a csv file
10. Implement EM clustering algorithm to cluster a set of data stored in a csv file
11. Implement perceptron classifier using python sklearn
12. Implement SVM classifier using python sklearn
13. Write a python program to do the performance analysis of classification algorithms on a specific dataset.